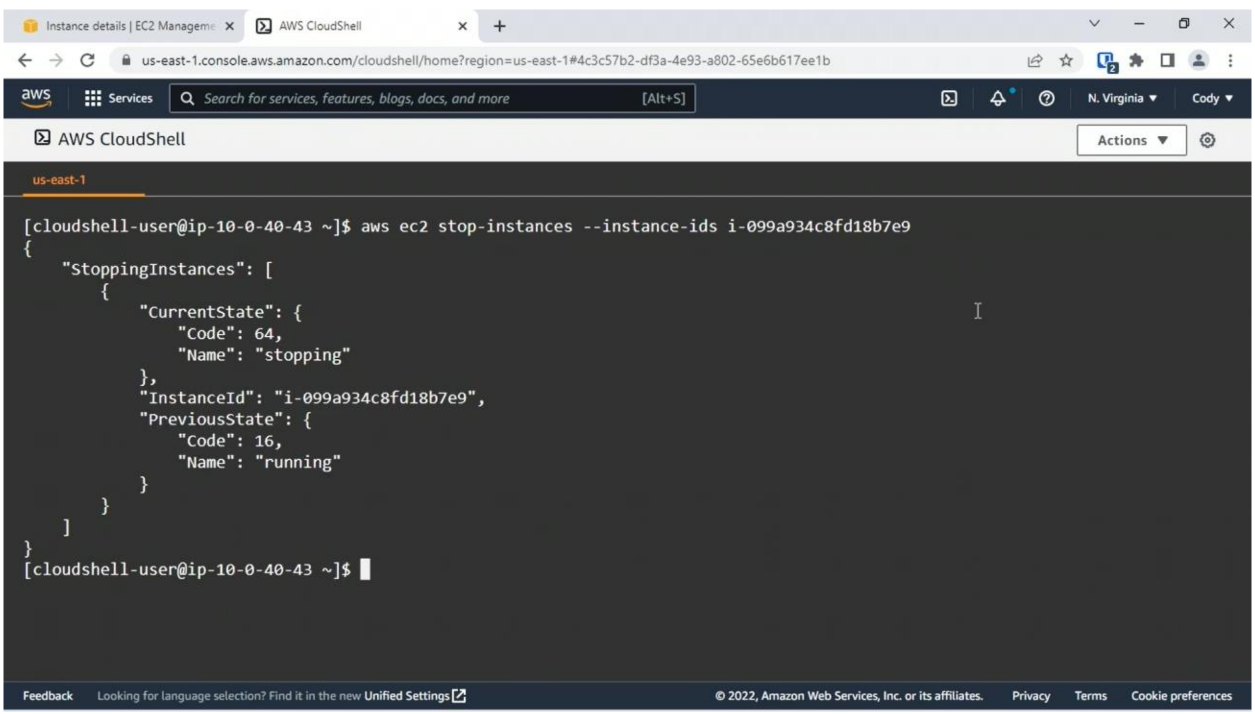
4.4 AWS Resource Scalability

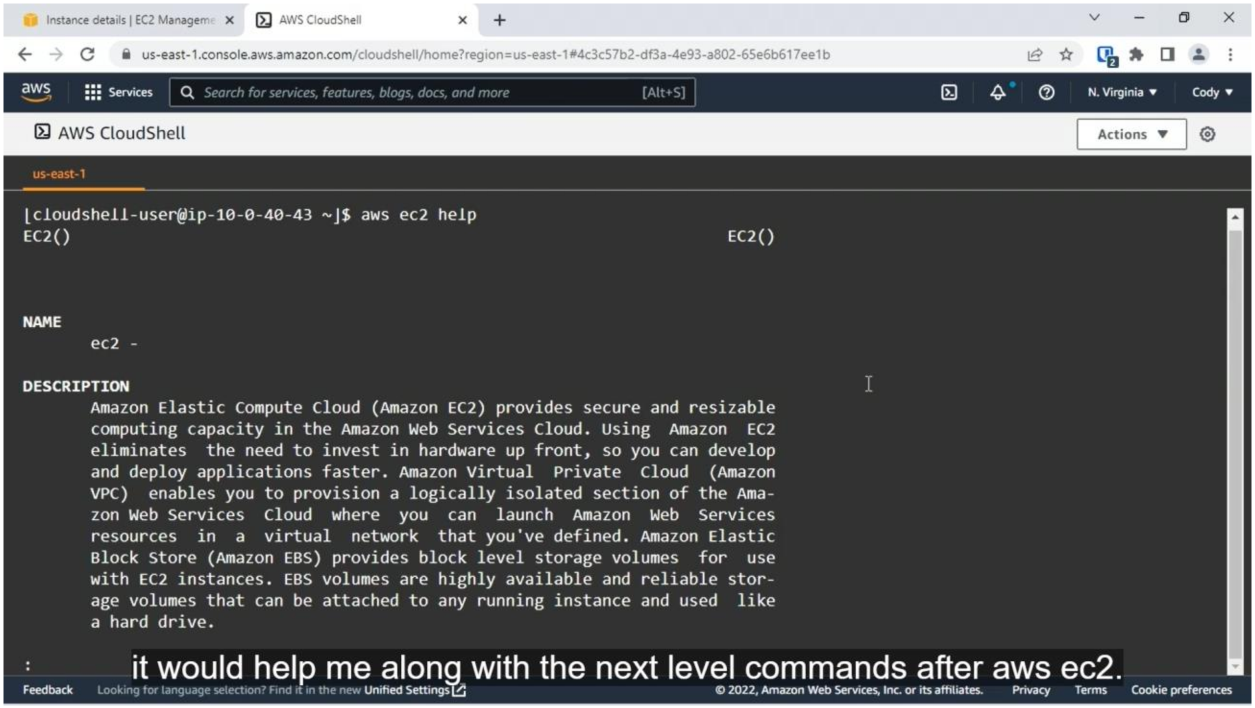
In this course, you will:

- Discover how to resize EC2 instances using the console and the CLI, followed by configuring auto-scaling to support an application workload
- Review when Amazon Fargate, EMR, and Batch should be used
- Discuss when various configurations of ElastiCache should be used
- Review AWS data analytics and ingestion services such as Amazon Athena and AWS Glue

Changing the EC2 Instance Sizing Using the Console

Changing the EC2 Instance Sizing Using the CLI




Instance details | EC2 Manageme
AWS CloudShell
us-east-1.console.aws.amazon.com/cloudshell/home?region=us-east-1#4c3c57b2-df3a-4e93-a802-65e6b617ee1b
Services
Search for services, features, blogs, docs, and more
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N. Virginia
Cody
AWS CloudShell
Actions
us-east-1
[cloudshell-user@ip-10-0-40-43 ~]$ aws ec2 help
EC2() EC2()
NAME
ec2 -
DESCRIPTION
Amazon Elastic Compute Cloud (Amazon EC2) provides secure and resizable computing capacity in the Amazon Web Services Cloud. Using Amazon EC2 eliminates the need to invest in hardware up front, so you can develop and deploy applications faster. Amazon Virtual Private Cloud (Amazon VPC) enables you to provision a logically isolated section of the Amazon Web Services Cloud where you can launch Amazon Web Services resources in a virtual network that you've defined. Amazon Elastic Block Store (Amazon EBS) provides block level storage volumes for use with EC2 instances. EBS volumes are highly available and reliable storage volumes that can be attached to any running instance and used like a hard drive.
:
it would help me along with the next level commands after aws ec2.
Feedback
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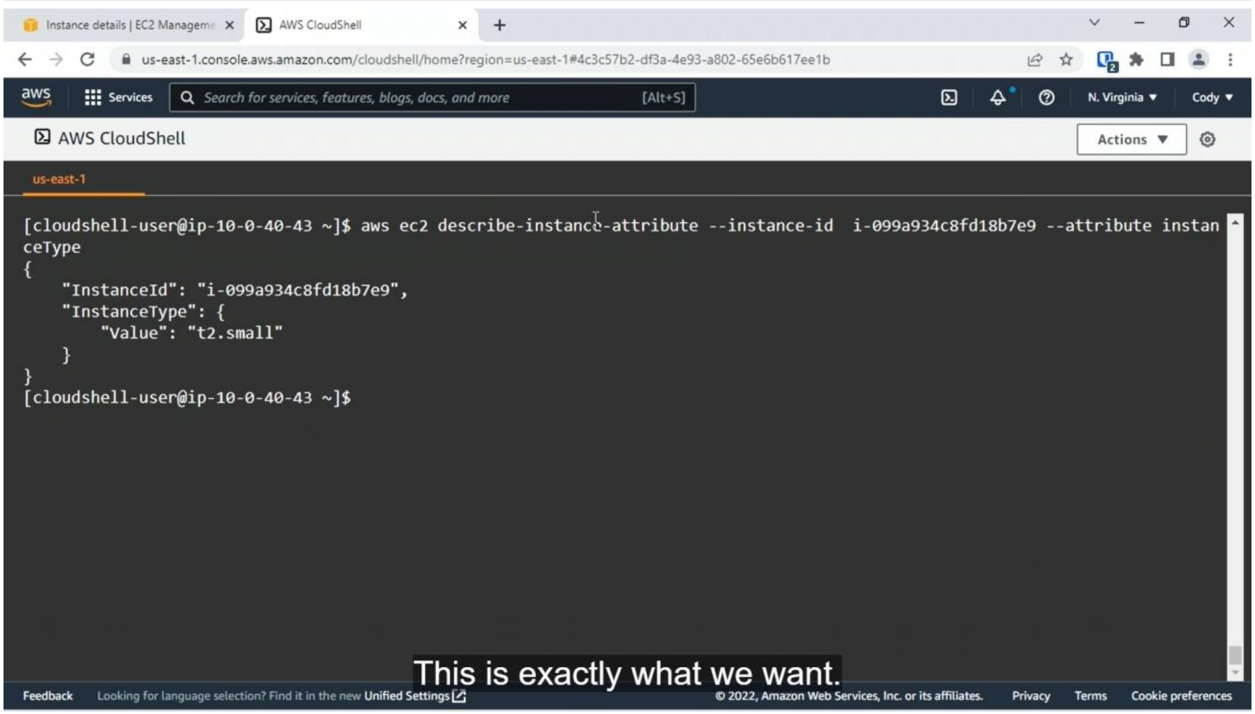
Instance details | EC2 Manageme
AWS CloudShell
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Services
Search for services, features, blogs, docs, and more
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Cody
AWS CloudShell
Actions
us-east-1
[cloudshell-user@ip-10-0-40-43 ~]$ aws ec2 describe-instance-attribute --instance-id i-099a934c8fd18b7e9 --attribute instanceType
{
"InstanceId": "i-099a934c8fd18b7e9",
"InstanceType": {
"Value": "t2.small"
}
}
[cloudshell-user@ip-10-0-40-43 ~]$
This is exactly what we want.
Feedback
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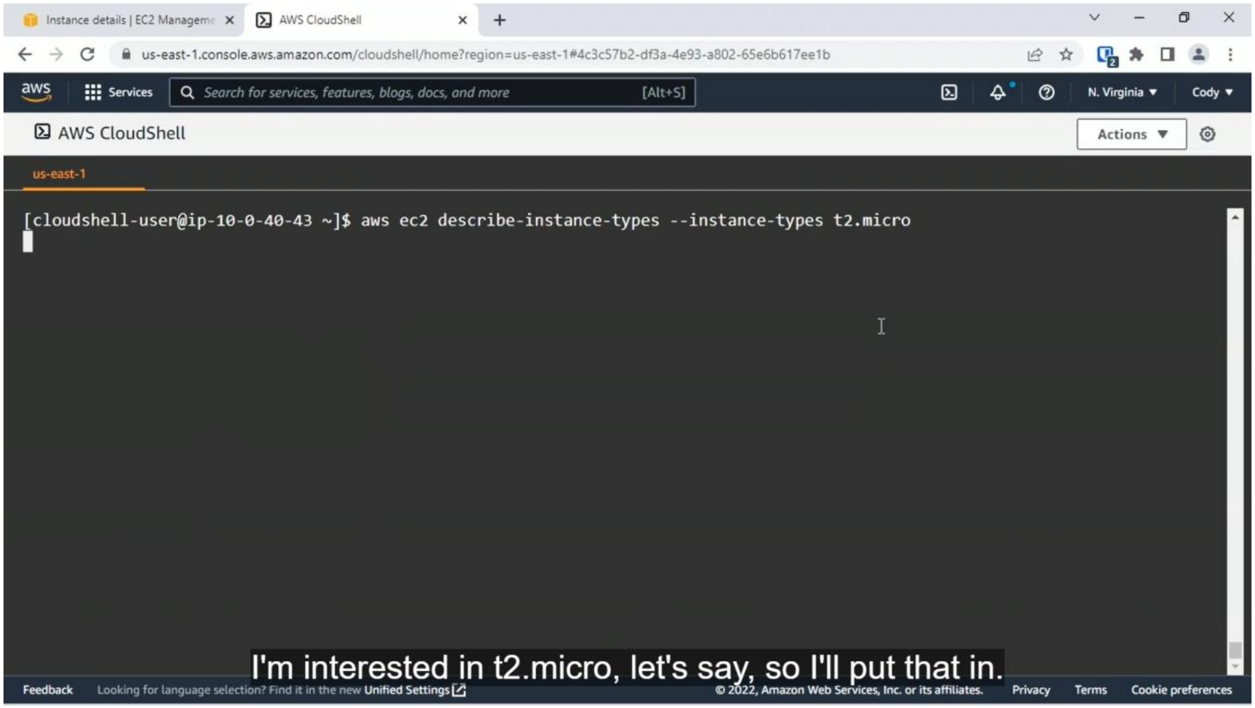
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aws
Services
Search for services, features, blogs, docs, and more
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Cody
AWS CloudShell
Actions
us-east-1
[cloudshell-user@ip-10-0-40-43 ~]$ aws ec2 describe-instance-types --instance-types t2.micro
I'm interested in t2.micro, let's say, so I'll put that in.
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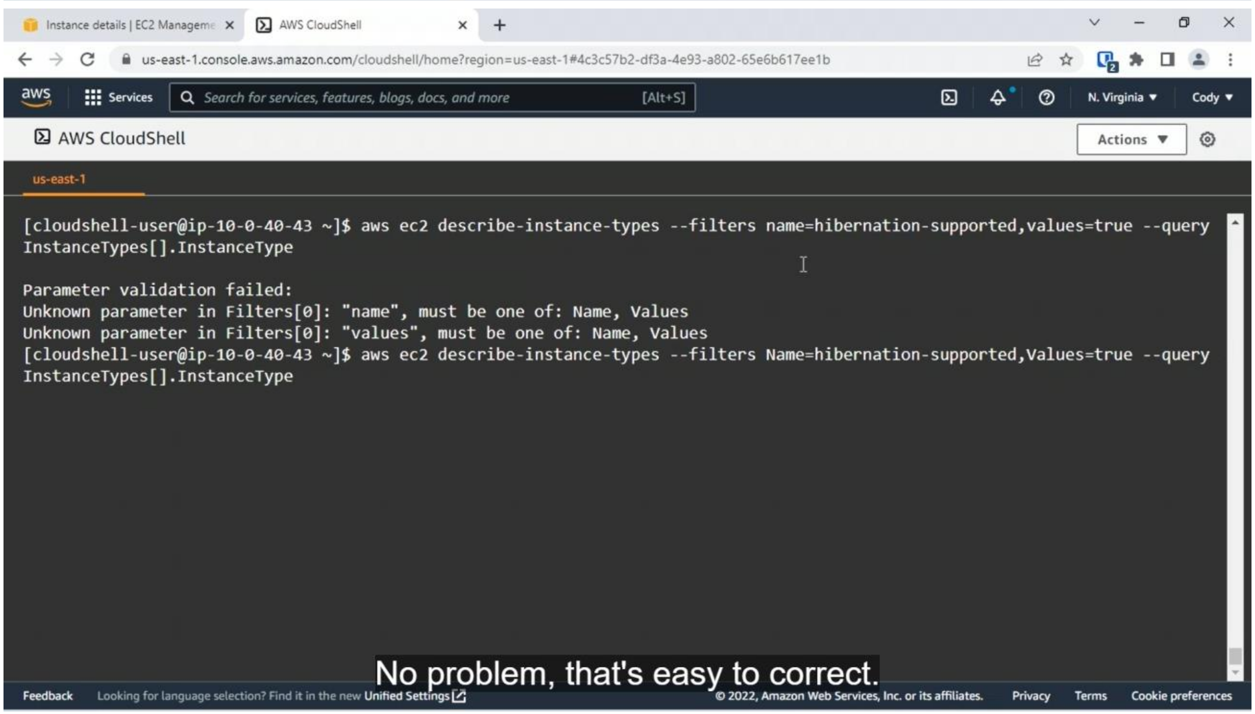
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Cody
AWS CloudShell
Actions
us-east-1
[cloudshell-user@ip-10-0-40-43 ~]$ aws ec2 describe-instance-types --filters name=hibernation-supported,values=true --query
InstanceTypes[].InstanceType
Parameter validation failed:
Unknown parameter in Filters[0]: "name", must be one of: Name, Values
Unknown parameter in Filters[0]: "values", must be one of: Name, Values
[cloudshell-user@ip-10-0-40-43 ~]$ aws ec2 describe-instance-types --filters Name=hibernation-supported,Values=true --query
InstanceTypes[].InstanceType
No problem, that's easy to correct.
Feedback
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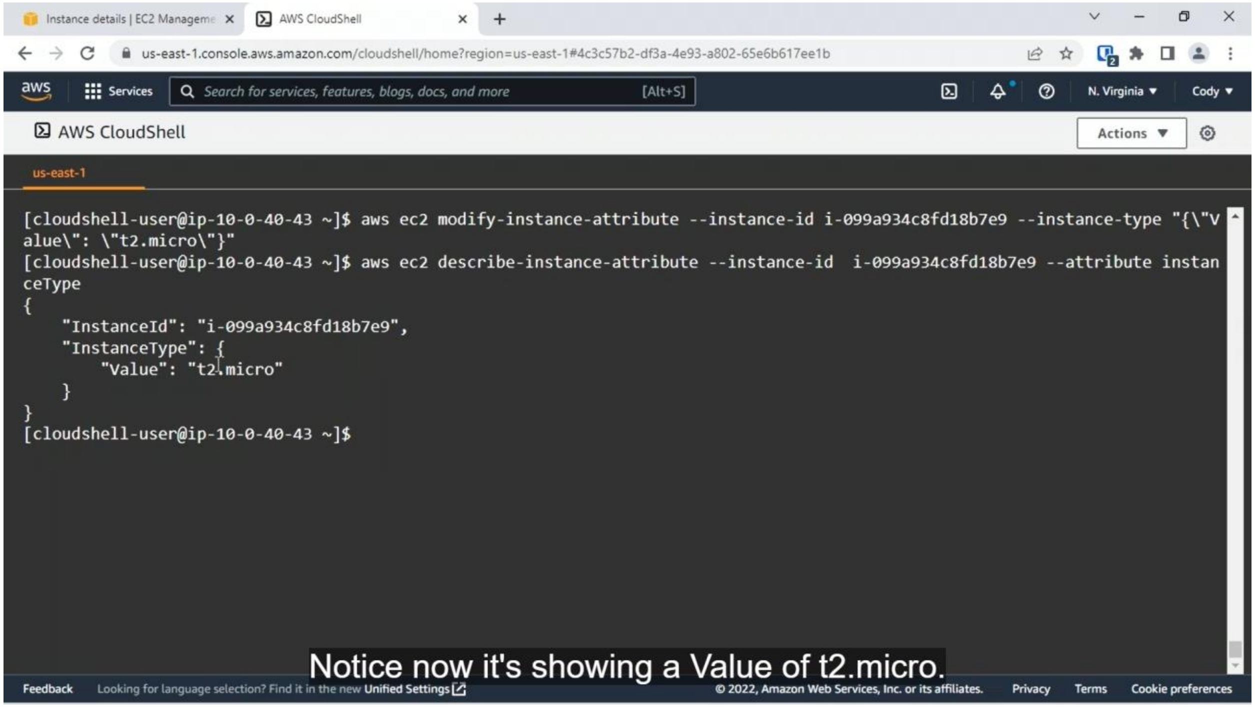
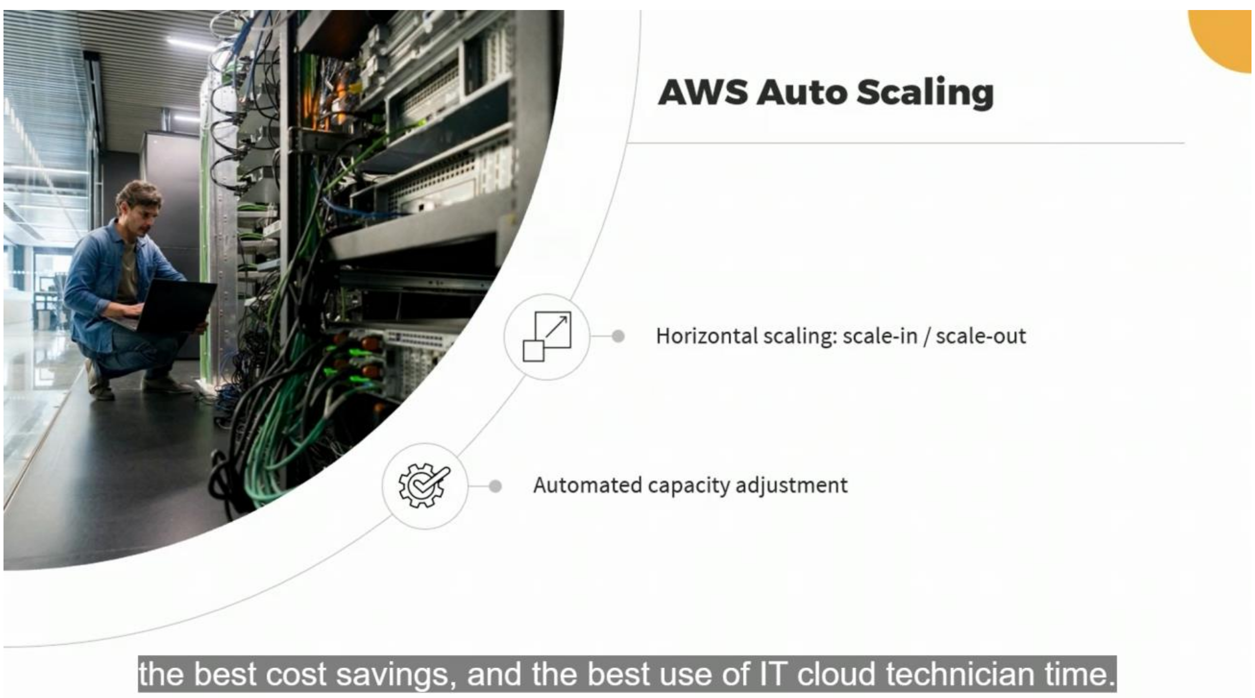


Auto Scaling



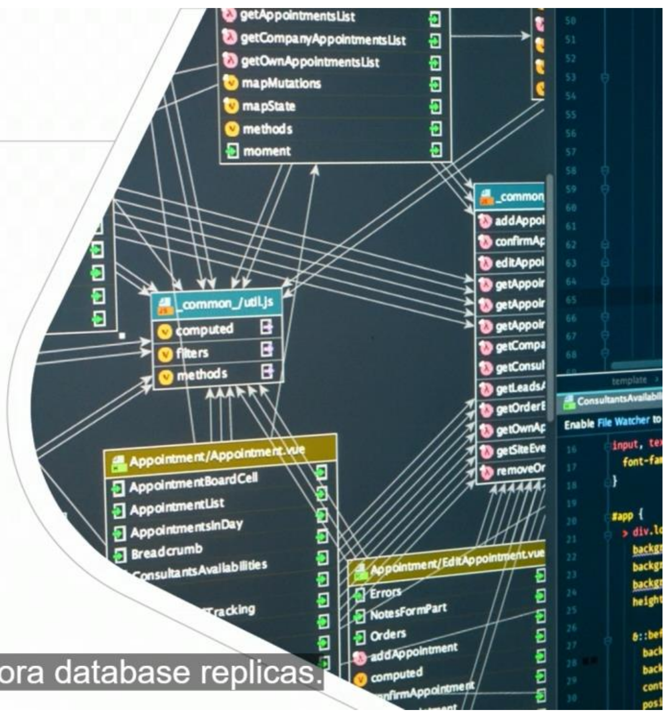
AWS Auto Scaling: Benefits

- App performance maintenance during busy peaks
- Scaling-in when app usage lessens, which costs less
- Use simple configuration and automated monitoring to help assess future capacity needs

You can monitor usage over time.

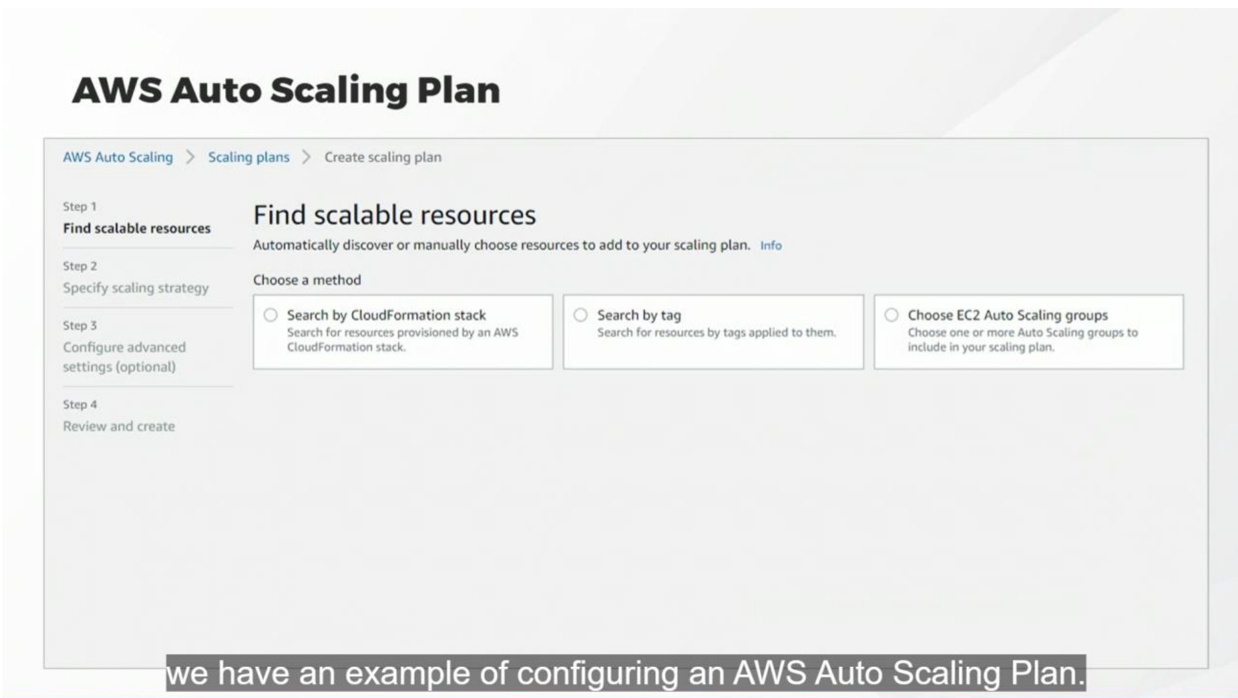
AWS Auto Scaling

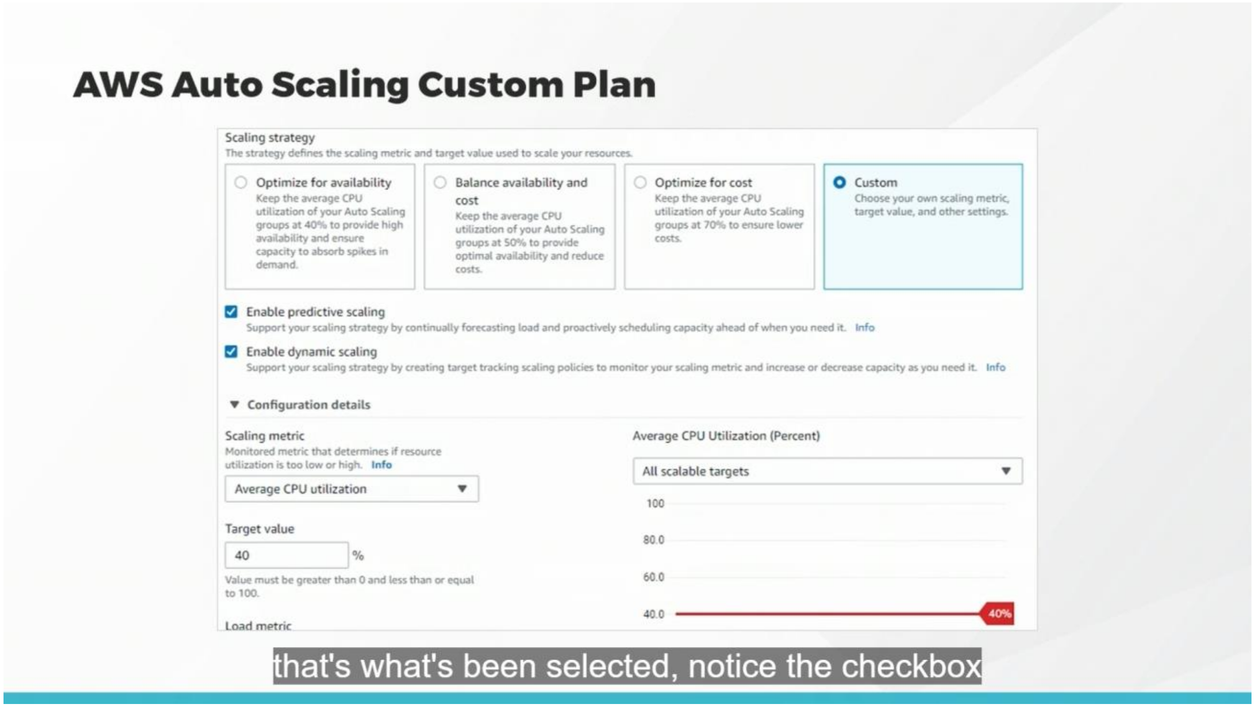
- EC2 instances
- Elastic Container Service (ECS)
- Amazon DynamoDB
- Amazon Aurora database replicas



and Amazon Aurora database replicas.

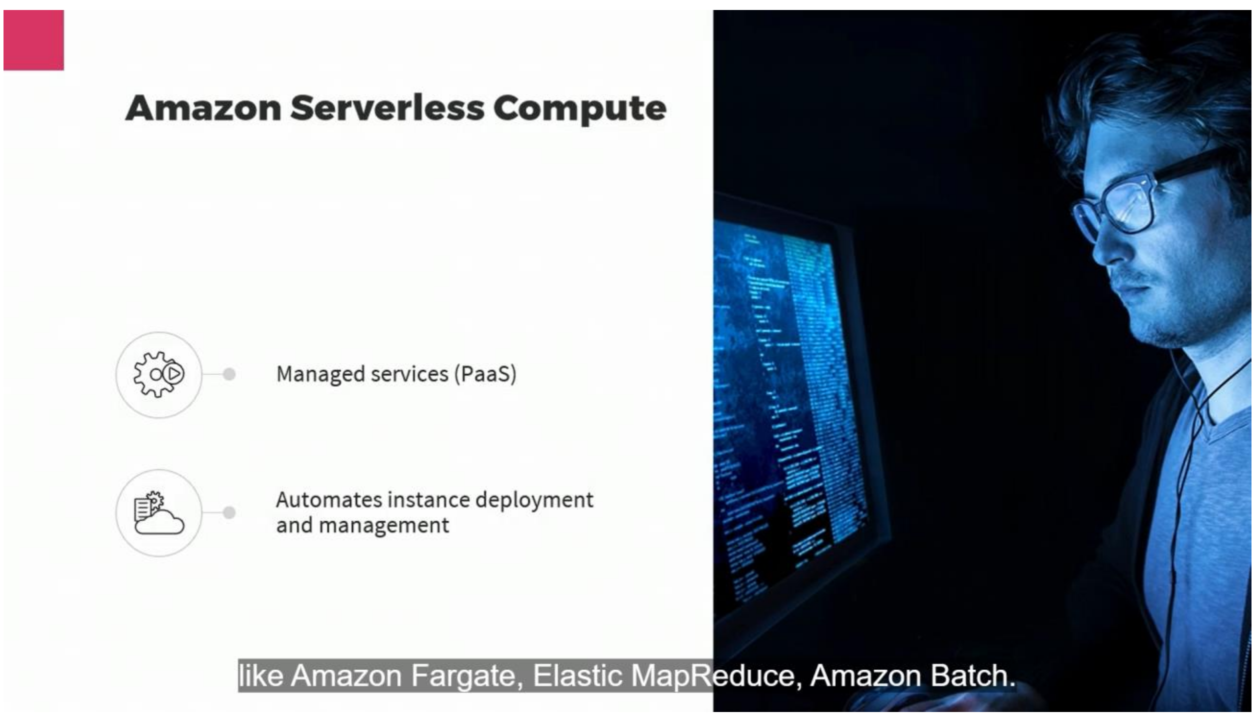

Predictive Scaling
Applies to EC2 Auto Scaling Groups
Scale proactively based on historical activity patterns
based on past historical activity patterns.


AWS Auto Scaling Plan
AWS Auto Scaling > Scaling plans > Create scaling plan
Step 1
Find scalable resources
Step 2
Specify scaling strategy
Step 3
Configure advanced settings (optional)
Step 4
Review and create
Find scalable resources
Automatically discover or manually choose resources to add to your scaling plan. Info
Choose a method
Search by CloudFormation stack
Search for resources provisioned by an AWS CloudFormation stack.
Search by tag
Search for resources by tags applied to them.
Choose EC2 Auto Scaling groups
Choose one or more Auto Scaling groups to include in your scaling plan.
we have an example of configuring an AWS Auto Scaling Plan.



Enabling Auto Scaling

AWS Serverless Compute



Amazon Batch

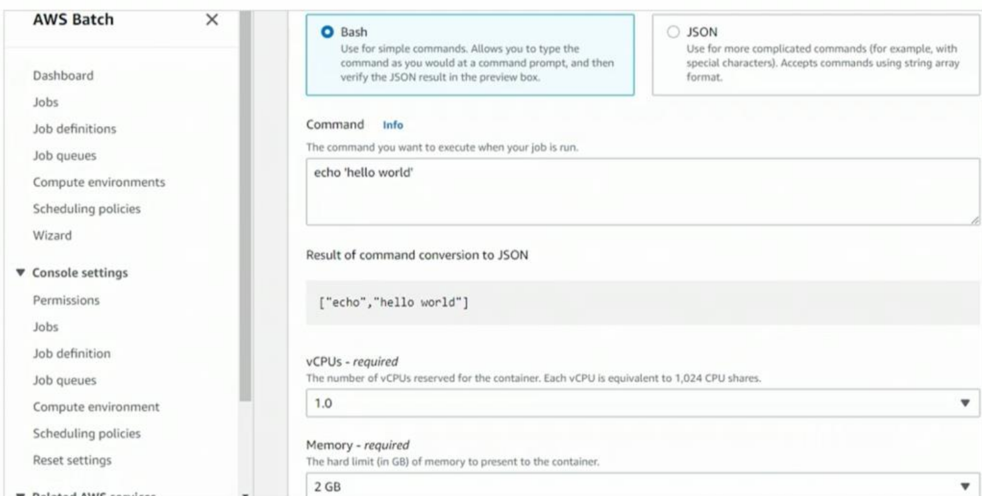
Batch job submission – data analytics, transformation, etc.

Compute and memory power – adjusted manually or automatically

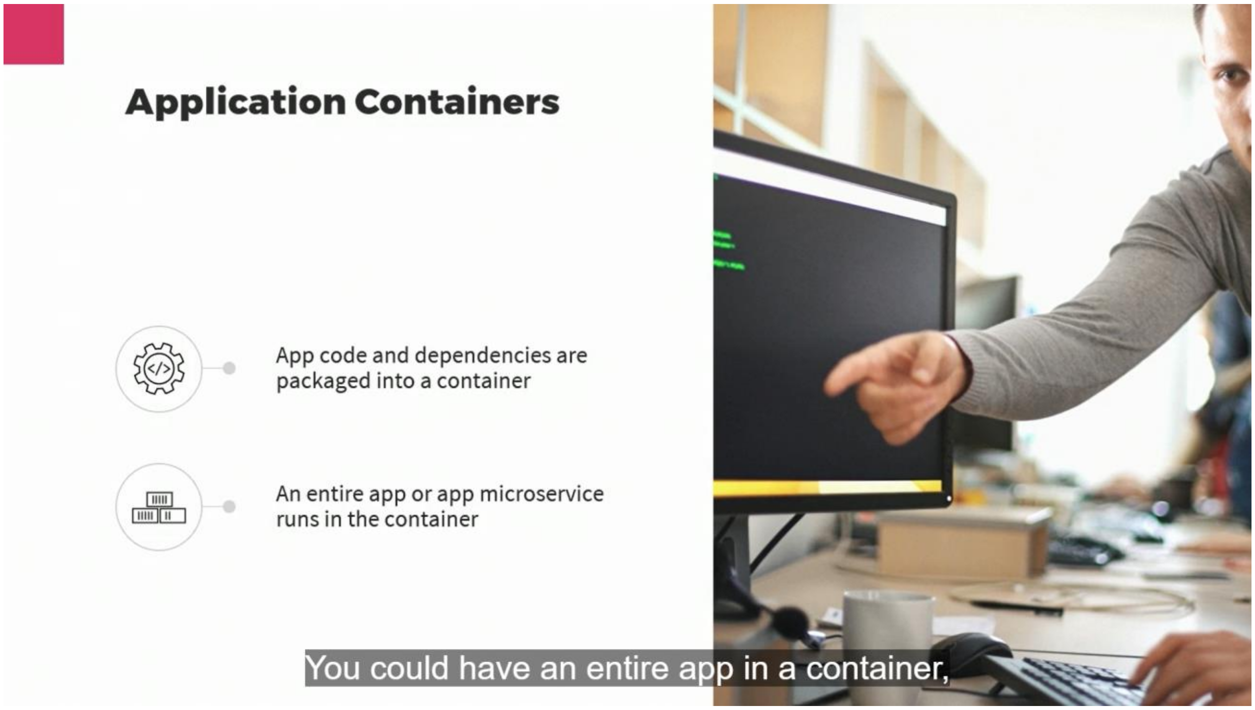
Jobs – scheduled to run on EC2, Spot Instances, or Amazon Fargate

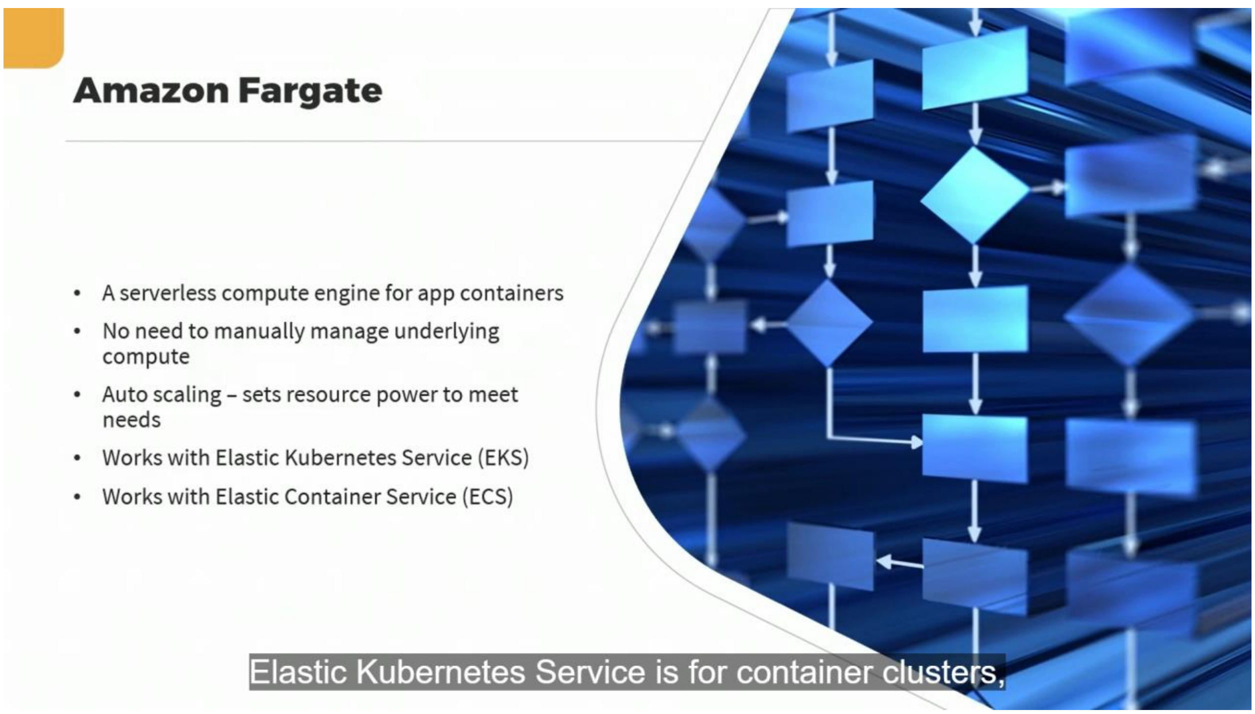
And you schedule the job to run on EC2 instances,

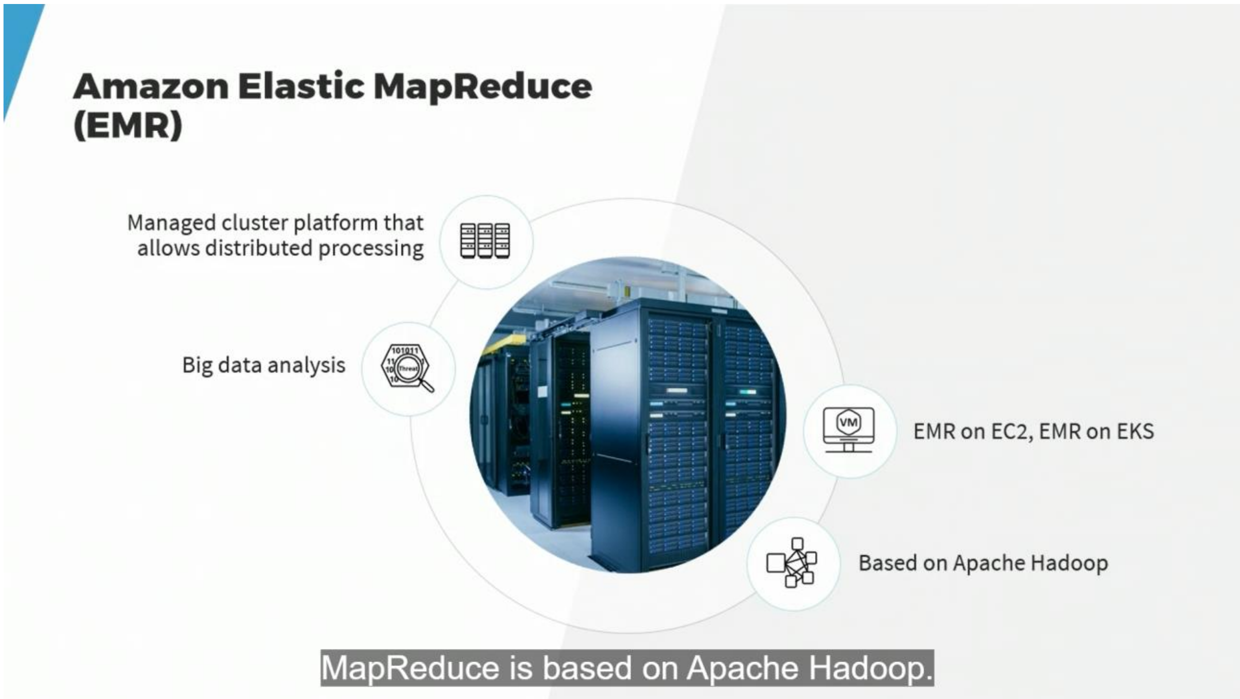
AWS Batch: Job Definition



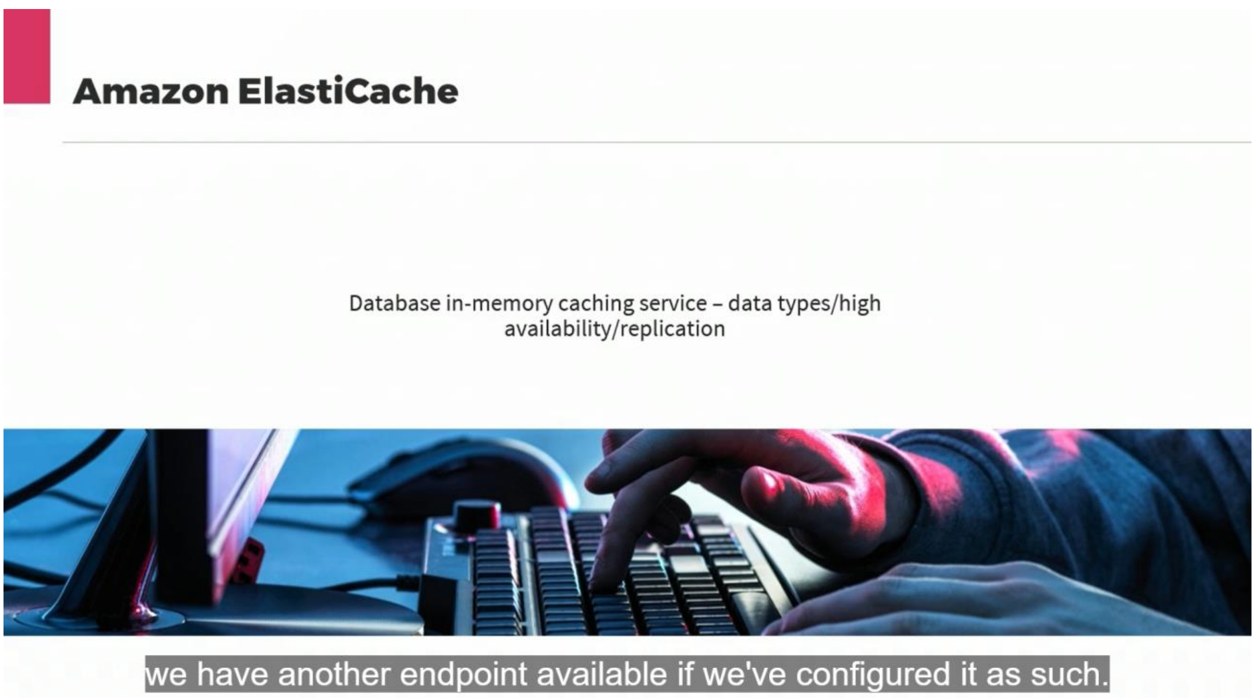
You can create a Bash job.

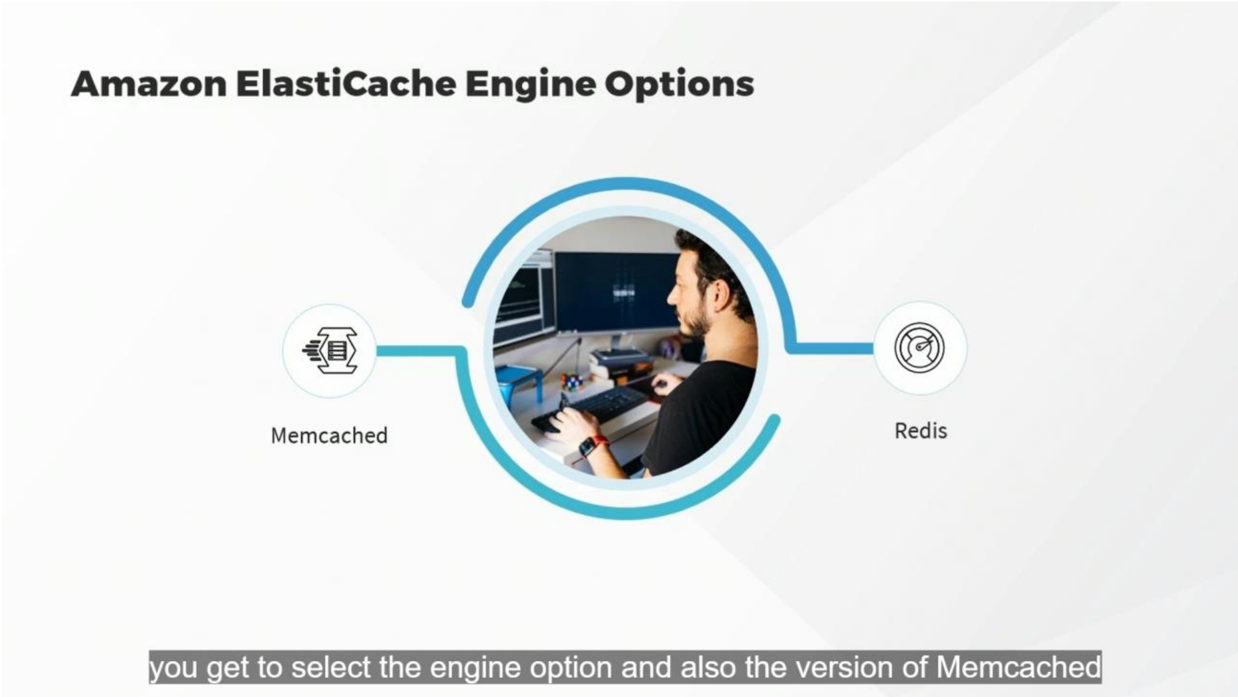

Application Containers
App code and dependencies are packaged into a container
An entire app or app microservice runs in the container
You could have an entire app in a container,

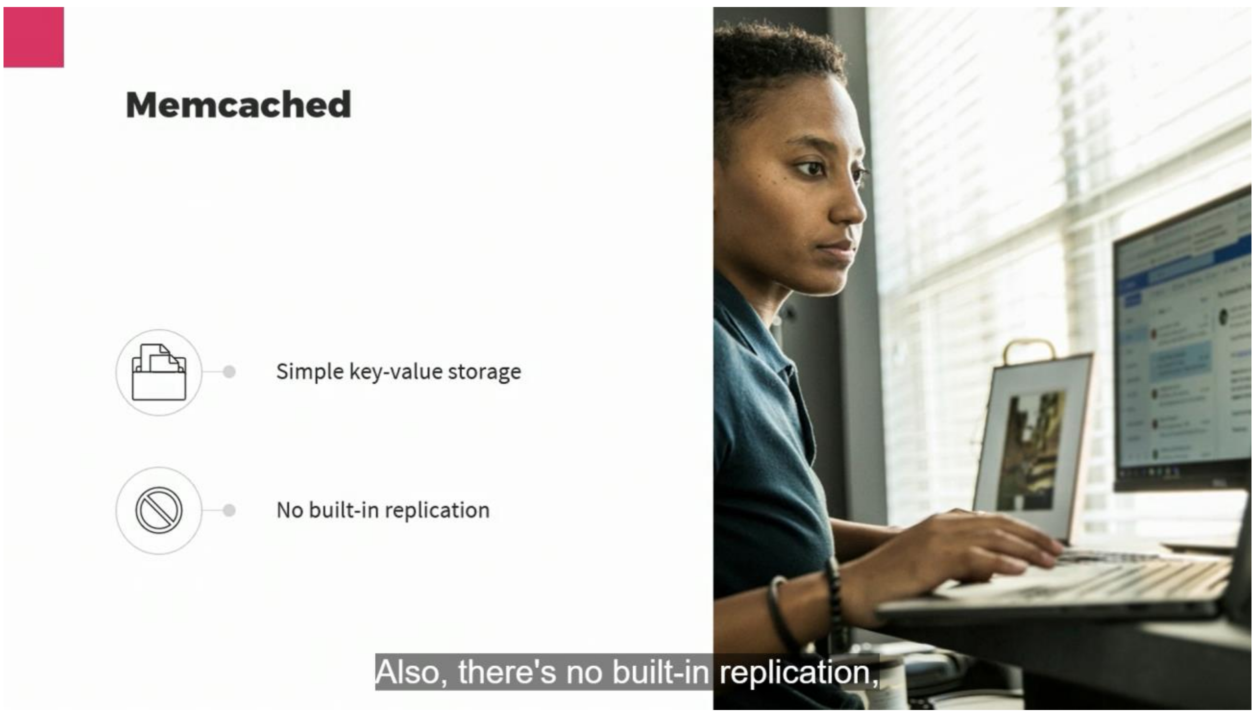

Amazon Fargate
A serverless compute engine for app containers
No need to manually manage underlying compute
Auto scaling – sets resource power to meet needs
Works with Elastic Kubernetes Service (EKS)
Works with Elastic Container Service (ECS)
Elastic Kubernetes Service is for container clusters,



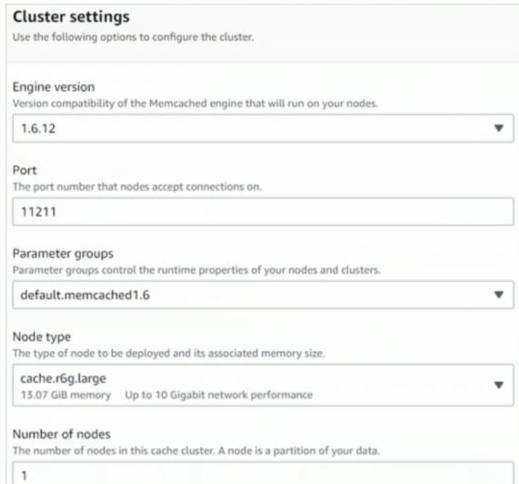
Amazon ElastiCache




Amazon ElastiCache Engine Options
Memcached
Redis
you get to select the engine option and also the version of Memcached


Memcached
Simple key-value storage
No built-in replication
Also, there's no built-in replication,

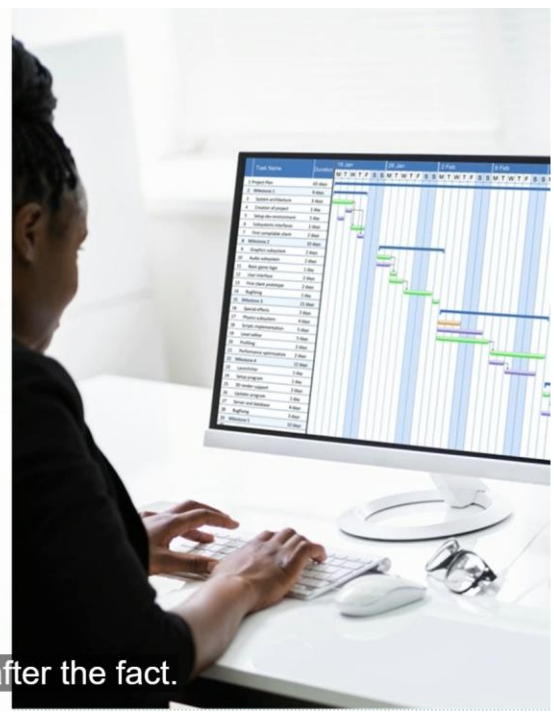
Memcached Cluster Settings



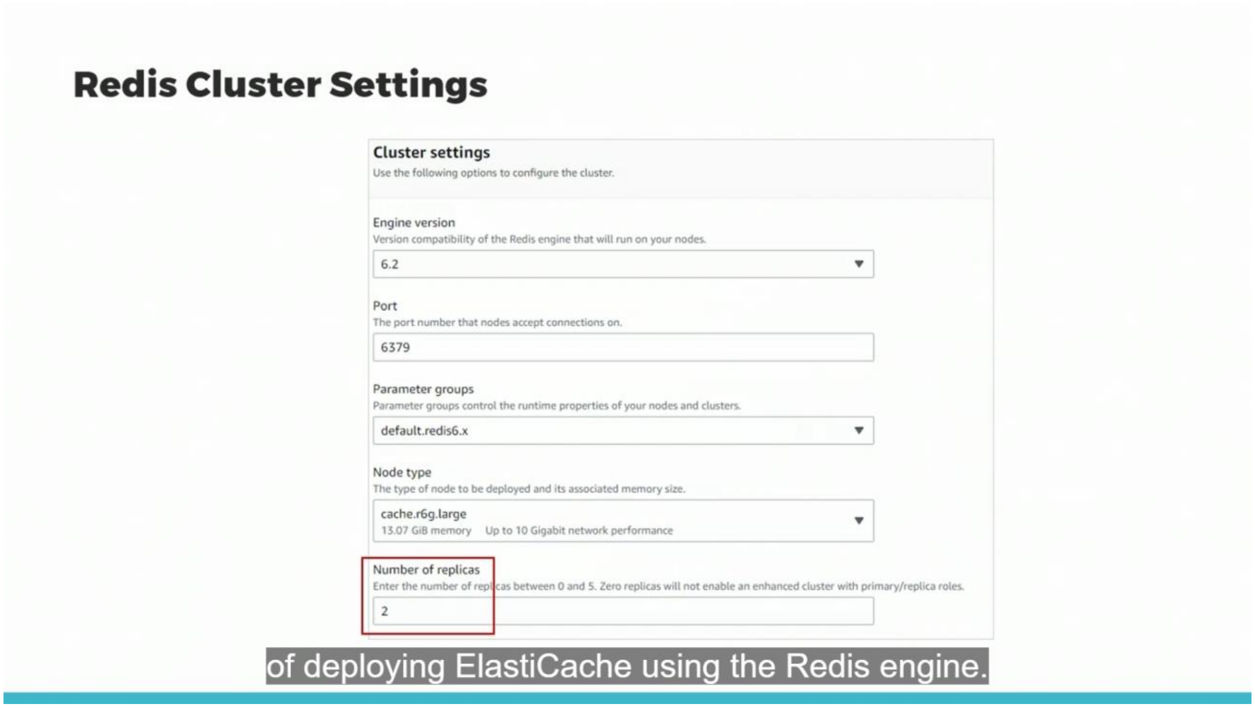
we have a screenshot of deploying a Memcached cluster.

Redis

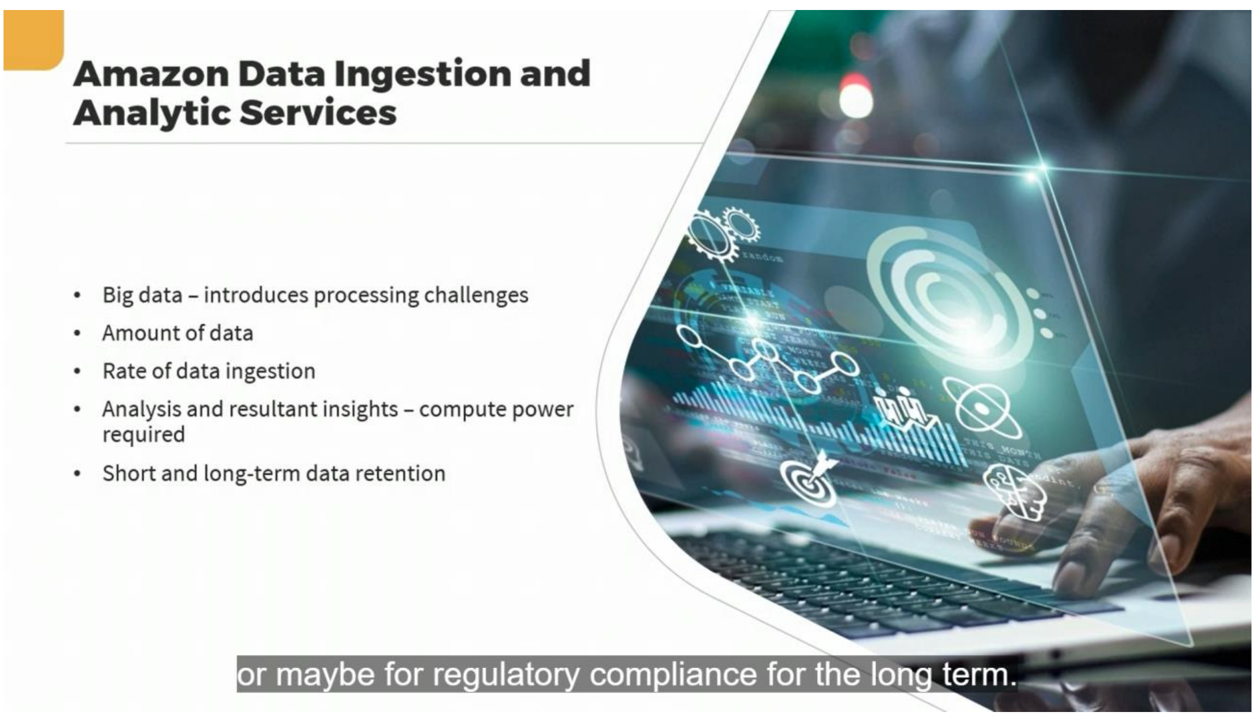
- Complex data types beyond value pairs (e.g., hashes, lists, bitmaps, etc.)
- Built-in replication, node failover, backup, and restore



You can change that after the fact.

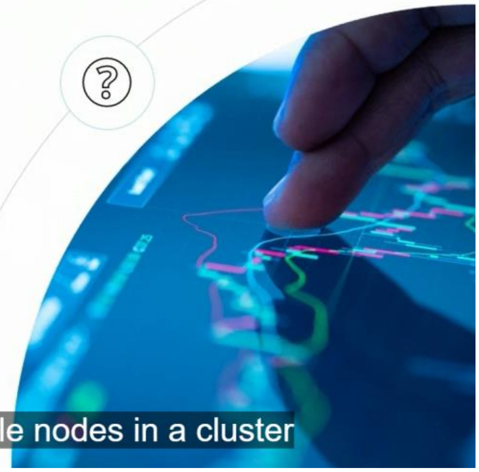


AWS Data Analytics and Ingestion Services



Amazon Athena

- Big data ANSI SQL querying service for data stored in S3
- Managed service
- Data – can be structured, semi-structured, or structureless
- Queries – executed in parallel across nodes for the utmost speed in query results



So at the exact same time across multiple nodes in a cluster

Amazon QuickSight

A Business Intelligence (BI) solution that results in data visualizations and dashboards

Is useful for forecasting trends and conducting what-if scenarios

It supports natural language queries against data from many different sources

And it supports standard querying against many different data sources.

Amazon Kinesis

Used to analyze and process streaming data

Processes data in real time instead of after collection

Is a managed service supporting automatic scaling

You don't have to deploy anything manually.

AWS Glue

Is a serverless extract, transform, load (ETL) service

Consists of a crawler, an ETL service, and a metadata data catalog

Can integrate with Amazon Athena

And it's one of those services that can integrate with Amazon Athena.

1. Which Elasticache engine is best suited to handle complex data types and replication?
    - Redis

2. To which cloud computing characteristic does auto scaling best apply?
    - Rapid elasticity

3. Which primary element is defined within an EC2 launch configuration?
    - AMI

4. You attempt to change the instance type for a Linux EC2 instance using the console, but the option is grayed out. What is the problem?
    - The instance is running

5. Which CLI command shows the current instance type for an EC2 instance?
    - aws ec2 describe-instance-attribute

6. You would like a managed solution that handles the underlying compute sizing for an EKS cluster. What should you use?
    - Fargate

7. Which Amazon service is designed as Business Intelligence (BI) solution?
    - QuickSight